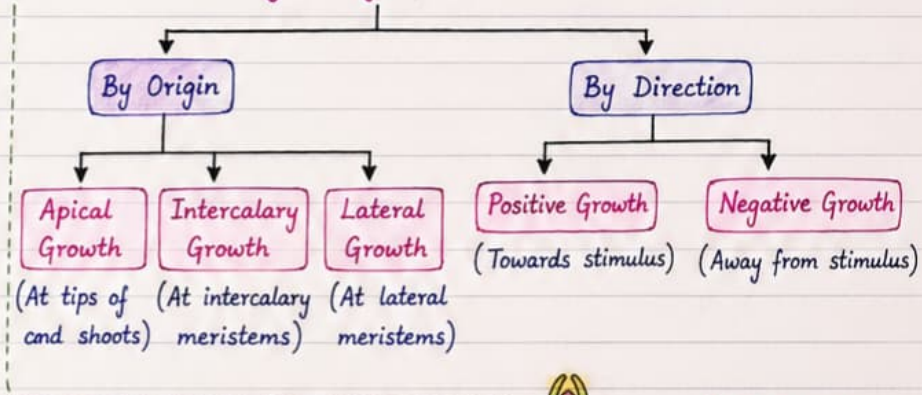


Growth : It is a permanent, irreversible increase in size, weight and volume of plant body due to cell division, cell enlargement and cell differentiation.

Key Points

- Irreversible & Permanent
- Quantitative
- Results in increase in dry weight
- Occurs in specific regions called meristems
- Regulated by hormones

Types of Growth

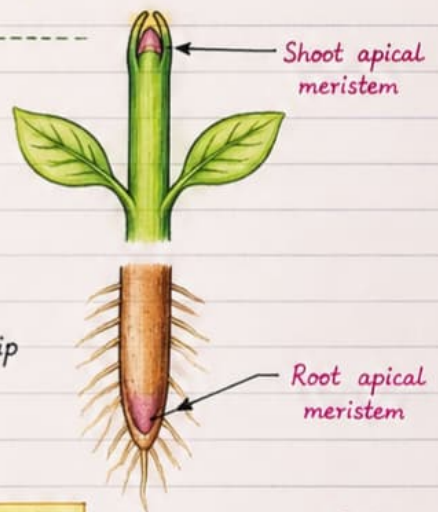


Meristems

- Meristems are regions of actively dividing cells.
- Types →
 1. Apical meristem
 2. Intercalary meristem
 3. Lateral meristem

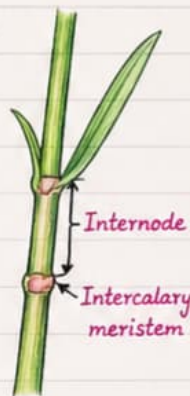
1. Apical Meristem

- Present at the tips of roots and shoots.
- Responsible for primary growth (increase in length).
- Example → Root tip, Shoot tip



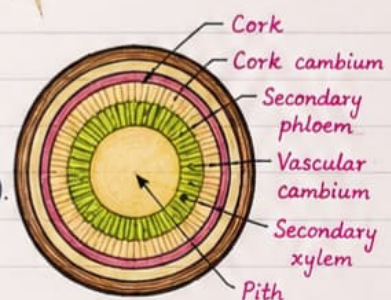
2. Intercalary Meristem

- Found at the base of leaves or at nodes.
- Helps in elongation of internodes.
- Common in grasses, e.g., wheat, sugarcane.



3. Lateral Meristem

- Present along the sides of stems and roots.
- Brings about secondary growth (increase in girth).
- Two types → Cambium and Cork cambium (Phellogen).



Phases of Growth

1. **Cell Division** → Increase in number of cells.
2. **Cell Enlargement** → Increase in size of cells.
3. **Cell Differentiation** → Cells acquire specific structure & function.

All three phases are necessary for overall growth.

Example

In a mango tree → Apical meristem increases its height, lateral meristem increases girth, and meristems produce new cells for leaves, branches and roots.



★ Remember → "Growth is quantitative, development is qualitative."

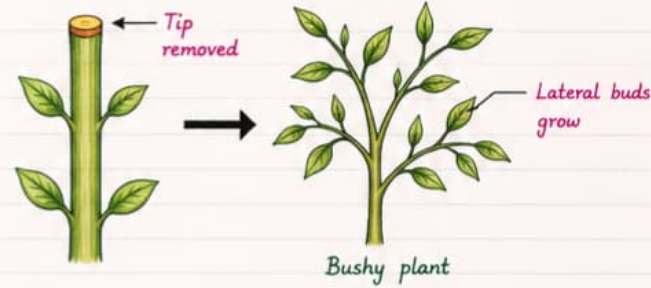
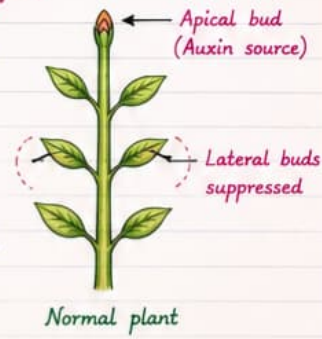
4. Role of Plant Hormones in Growth and Development

- Plant hormones are chemical messengers that regulate various growth processes.

Hormone	Source	Functions
Auxins	Shoot tips, young leaves, seeds	- Cell elongation - Apical dominance - Root initiation - Tropic movements
Gibberellins	Young leaves, seeds, fruits	- Stem elongation - Breaking seed dormancy - Bolting in rosette plants - Fruit growth
Cytokinins	Root tips, young fruits, seeds	- Cell division - Shoot initiation - Delays leaf senescence
Ethylene	Ripening fruits, leaves, stems	- Fruit ripening - Leaf fall, senescence - Tripple response in seedlings
Abscissic Acid (ABA)	Leaves, seeds, fruits	- Inhibits growth - Induces seed dormancy - Stomatal closure (stress conditions)

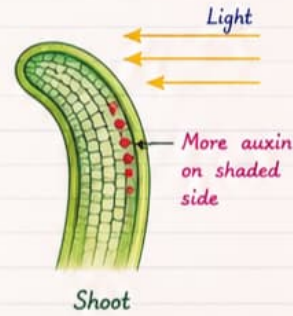
★ Apical Dominance (Auxin Effect)

- Auxin produced at the tip suppresses the growth of lateral buds.
- Removal of shoot tip (decapitation) → lateral buds grow → bushy plant.



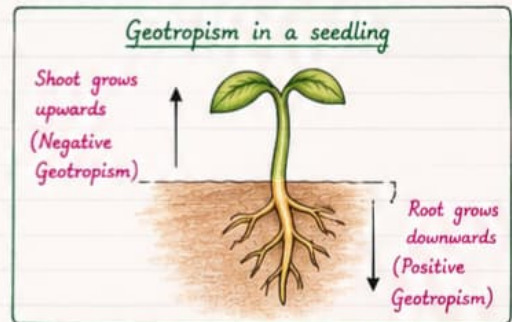
★ Phototropism (Auxin Role)

- When light comes from one side, auxin migrates to the shaded side.
- Cells on shaded side elongate more.
- Shoot bends towards light.



5. Other Growth Movements

- Geotropism (Gravitropism)** → Growth in response to gravity.
e.g., Roots → Positive, Shoots → Negative
- Hydrotropism** → Growth in response to water.
e.g., Roots grow towards water.
- Chemotropism** → Growth in response to chemicals.
e.g., Pollen tube grows towards ovule due to chemicals.



6. Nastic Movements

These movements are independent of the direction of stimulus and are caused by changes in turgor pressure.

Thigmonasty (Touch)	Nyctinasty (Day-Night)	Seismonasty (Shock/Vibration)	Photonasty (Light)
e.g., Mimosa pudica (Leaves fold on touch)	e.g., Cassia, Oxalis (Leaves fold at night)	e.g., Mimosa (Leaves fold on shock)	e.g., Sunflower (Flowers face sunlight)

Key Takeaways

- ✓ Growth is permanent & irreversible.
- ✓ Meristems are responsible for growth.
- ✓ Growth can be by origin or direction.
- ✓ Plant hormones regulate growth processes.
- ✓ Plants show various movements in response to stimuli.

★ Remember → "From a tiny seed to a giant tree, it's all about Growth and Development."

7. Plant Growth Regulators (PGRs)

→ PGRs are natural or synthetic organic compounds (other than nutrients) that, in low concentration, regulate various aspects of plant growth and development.

★ They may promote, inhibit or modify plant growth processes.

PGR	Natural Source	Major Functions	Examples / Uses
Auxins	IAA (Indole acetic acid) in shoot tips, young leaves, fruits	- Promote cell elongation - Apical dominance - Root initiation - Tropic responses	- IAA, NAA, 2,4-D - Used in rooting powders, weed control, fruit set (parthenocarp)
Gibberellins	Young leaves, seeds, fruits, embryos	- Stem elongation - Breaking seed dormancy - Bolting, flowering (in some) - Fruit growth	- GA₃ (Gibberellic acid) - Used in malting of barley, increasing grape size, breaking dormancy
Cytokinins	Root tips, developing seeds, fruits	- Cell division - Shoot initiation - Delay leaf senescence	- Kinetin, BAP - Used in tissue culture (shoot multiplication)
Ethylene	Ripening fruits, leaves, stems, ageing tissues	- Fruit ripening - Leaf / flower abscission - Senescence	- Ethene (C₂H₄) - Used for artificial ripening of fruits
Absciscic Acid (ABA)	Leaves, seeds, fruits	- Inhibits growth - Induces seed dormancy - Stomatal closure (stress)	- ABA - Used in stress tolerance studies, dormancy control

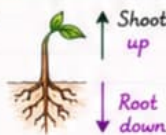
8. Plant Responses to Stimuli

Plants respond to their environment for survival.

- Phototropism → Growth in response to light.
e.g., Shoot bends towards light.



- Geotropism (Gravitropism) → Growth in response to gravity.
e.g., Roots grow downward (positive), Shoots grow upward (negative).



- Hydrotropism → Growth in response to water.
e.g., Roots grow towards water.



- Thigmotropism → Growth in response to touch.
e.g., Tendrils coil around support.



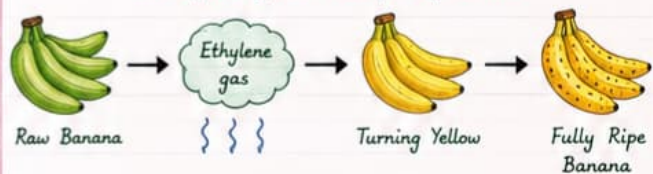
- Chemotropism → Growth in response to chemicals.
e.g., Pollen tube grows towards ovule.



9. Application of PGRs

- Auxins (2,4-D) → Weed control (kills dicot weeds)
- Auxins (NAA, IBA) → Rooting of cuttings.
- Gibberellins → Breaking seed dormancy, increasing size of grapes, malting.
- Cytokinins → Used in tissue culture for shoot multiplication.
- Ethylene → Ripening of fruits like banana, mango, tomato.
- ABA → Inducing dormancy, closing stomata during water stress.

Use of Ethylene in Ripening Fruits



10. Significance of Growth and Development

- ✓ Increase in size and weight leads to more biomass.
- ✓ Ensures proper formation of organs, tissues and systems.
- ✓ Helps in reproduction (flowering, fruiting, seed formation).



- ✓ Helps plants adapt to environment and survive.
- ✓ Basis of agriculture, horticulture and forestry.
- ✓ Essential for maintaining life cycle and continuity of species.

Remember!

Growth is increase in number/size.
Development is qualitative change leading to maturity.

★ Growth gives us quantity, Development gives us quality and maturity.  

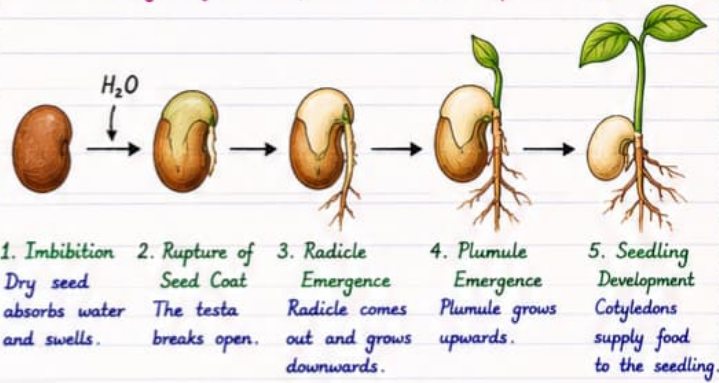
11. Seed Germination

Germination is the process by which a seed resumes growth and develops into a new plant.

Conditions Required

- **Water (Moisture)** → Activates enzymes and softens seed coat.
- **Oxygen** → Required for respiration.
- **Suitable Temperature** → Optimum temperature helps enzyme activity.
- **Light** → Required for some seeds.

Stages of Seed Germination (Example: Bean)



Types of Germination

Epigeal Germination

- Cotyledons come above the soil.
- Hypocotyl elongates.
- **Example** → Bean, Castor, Sunflower.



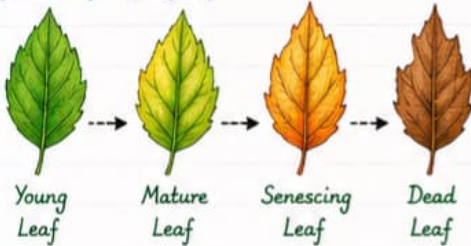
Hypogeal Germination

- Cotyledons remain below the soil.
- Epicotyl elongates.
- **Example** → Pea, Gram, Maize.



12. Ageing and Senescence

Ageing is a natural, gradual process. Senescence is the final stage of ageing.



Features of Senescence

- Degradation of chlorophyll → yellowing of leaves.
- Transport of food substances to storage organs.
- Decrease in respiration and metabolic activities.
- Cells lose their ability to divide.
- Finally, tissues and organs die.

Example

Shedding of leaves in autumn, drying of petals and leaves.

Significance

- ★ Removes worn out parts.
- ★ Makes way for new growth.
- ★ Recycling of nutrients to other parts.
- ★ Helps in survival of plant.

13. Importance of Plant Growth and Development

Growth and development are essential for the life of plants and for the ecosystem.

Food Production	Raw Materials	Oxygen Supply	Environmental Balance	Beautification	Support Life
					
Growth of crops provides food for humans and animals.	Plants provide wood, fibres, rubber, gums, oils and medicines.	Through photosynthesis, plants release oxygen essential for life.	Plants maintain ecological balance, soil fertility and water cycle.	Plants add beauty and improve mental well-being.	Plants are the base of all food chains.

SUMMARY

- Growth is an irreversible increase in size, weight and volume.
- Development involves qualitative changes leading to maturity.
- Growth occurs by cell division, enlargement and differentiation.
- Plant hormones (PGRs) regulate and coordinate growth processes.
- Plants respond to stimuli through various movements.
- Seed germination requires water, oxygen, suitable temperature and sometimes light.
- Senescence is the last stage of ageing and is beneficial.
- Growth and development are vital for survival and usefulness of plants.

Remember!

- ✓ From a tiny seed to a giant tree, it's all about Growth and Development.
- ✓ Growth is quantitative, Development is qualitative.
- ✓ Hormones are the messengers, Plants are the responders!